

SHOWTIME OTT PLATFORM

LINEAR REGRESSION ANALYSIS

Predicting First-Day Content Viewership

A Project Report on Exploratory Data Analysis, Data Preprocessing,
Linear Regression Modelling, Assumption Testing and Business Recommendations

Submitted as part of the Linear Regression coursework

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Project Overview

This report presents a structured analysis of first-day content viewership on the ShowTime OTT platform, covering exploratory data analysis, data preprocessing, multiple linear regression modelling, assumption testing, and model performance evaluation. Each section states the business question, applies the appropriate statistical or modelling technique, and translates the results into a clear, actionable business conclusion.

Section	Focus	Method / Approach
1	Introduction	Business context, objectives, dataset overview
2	Exploratory Data Analysis	Univariate, bivariate and multivariate analysis
3	Data Preprocessing	Feature engineering, encoding, train-test split
4	Model Building	Multiple Linear Regression (OLS)
5	Assumption Testing	Linearity, independence, normality, homoscedasticity, multicollinearity, influence
6	Model Performance	MAE, RMSE, MAPE, R-squared (train vs test)
7	Business Insights	Key drivers of viewership and recommendations
8	Conclusion	Summary, limitations, executive recap

Executive Summary

Objective

Identify the factors driving first-day content viewership on the ShowTime OTT platform and build a model to forecast first-day views for upcoming releases.

Approach

Multiple linear regression was fitted on 1,000 content releases, using platform traffic, advertising impressions, trailer engagement, sporting-event overlap, genre, release day, and season as predictors. The model was trained on 800 releases and validated on 200 held-out releases.

Model Performance

The model explains 77.4% of the variation in first-day viewership on unseen data (training $R^2 = 0.787$, test $R^2 = 0.774$), with a test-set mean absolute error of 0.0399 million views. All five regression assumptions were tested and satisfied.

Key Findings

Factor	Effect on First-Day Views	Significance
Trailer views	+2,311 views per million trailer views	$p < 0.001$
Major sporting event	-59,600 views (-12.6% of average)	$p < 0.001$
Saturday vs Friday release	+52,600 views (+11.1%)	$p < 0.001$
Wednesday vs Friday release	+49,500 views (+10.5%)	$p < 0.001$
Summer vs Fall release	+44,600 views (+9.4%)	$p < 0.001$
Platform visitors	+12,900 views per million visitors	$p < 0.001$
Advertising impressions	No measurable effect	$p = 0.180$
Genre	No measurable effect	$p \geq 0.14$ (all)

Headline Conclusion

ShowTime releases 37% of its content on Friday — its worst-performing day of the week.

Friday averages 0.447 million first-day views against Saturday's 0.498 million and Wednesday's 0.495 million. Rebalancing the release calendar, avoiding sporting-event clashes, and favouring Summer release windows together represent an estimated **33% improvement in first-day viewership**, with no additional content or marketing spend required.

Meanwhile, advertising impressions — which span a two-fold range across releases — show no detectable relationship with first-day viewership, while trailer engagement emerges as by far the strongest predictor available before a title even launches.

Section 1 — Introduction

1.1 Business Context

ShowTime is an OTT media service provider offering movies, web shows, and other digital content to its users. The platform wants to understand the factors that influence the first-day viewership of newly released content.

The first-day viewership of content can be influenced by several factors, such as the number of visitors to the platform, advertising and promotional activities, trailer engagement, release day, season, genre, and the occurrence of major sporting events.

1.2 Business Objective

The objective of this analysis is to identify the key factors that influence first-day content viewership and develop a Linear Regression model that can predict the first-day views of content on the ShowTime platform.

Target Variable: views_content — First-day content views, in millions

1.3 Key Questions to be Answered

The following questions are specified for the EDA section of this project:

- What does the distribution of content views look like?
- What does the distribution of genres look like?
- The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?
- How does the viewership vary with the season of release?
- What is the correlation between trailer views and content views?

All variables in the dataset are examined thoroughly in the EDA section beyond these five questions, as instructed, so that every predictor's relationship with viewership — not only the ones named above — is understood before modelling.

1.4 Dataset Overview

The dataset contains 1,000 observations and 8 variables related to content releases on the OTT platform. The variables include information about platform visitors, advertising impressions, major sports events, content genre, release day, release season, trailer views, and first-day content views.

1.5 Variable Classification

Variable	Type	Description
visitors	Numerical	Average platform visitors in the past week, in millions
ad_impressions	Numerical	Advertising impressions for the content, in millions

Variable	Type	Description
major_sports_event	Categorical / Binary	Whether a major sports event occurred on the release day
genre	Categorical	Genre of the content
dayofweek	Categorical	Day on which the content was released
season	Categorical	Season in which the content was released
views_trailer	Numerical	Trailer views, in millions
views_content	Numerical	First-day content views, in millions (target)

Table 1: Variable Classification

Section 2 — Exploratory Data Analysis

2.1 Data Quality Check

Before analysis, the dataset was checked for structural issues: shape, data types, duplicate rows, and missing values.

Shape: 1,000 rows × 8 columns

Duplicate rows: 0

Missing values: 0 (across all columns)

The dataset contains no duplicate records and no missing values. No duplicate removal or missing-value imputation is required.

2.2 Distribution of Content Views (Question 1)

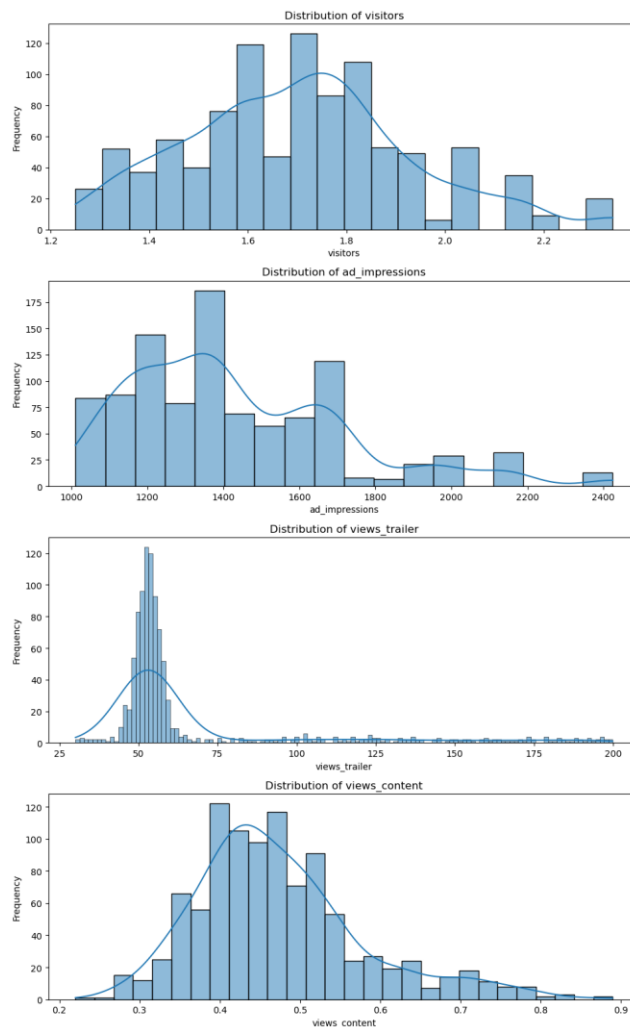


Figure 1: Distribution of Numerical Variables

Variable	Skewness
visitors	0.37
ad_impressions	1.03
views_trailer	2.37
views_content	0.94

Answer — What does the distribution of content views look like?

First-day content viewership (views_content) has a mean of 0.473 million views, a median of 0.45 million, and ranges from 0.22 to 0.89 million. The distribution is positively skewed (skewness = 0.94): most releases achieve between roughly 0.35 and 0.55 million views, with a smaller number of releases reaching substantially higher first-day viewership. This right tail represents genuine high-performing content rather than a symmetric spread around the mean.

For context, visitors is mildly skewed (0.37), ad_impressions moderately skewed (1.03), and views_trailer is the most heavily skewed variable in the dataset (2.37) — its distribution is examined further since it turns out to be the strongest driver of viewership.

2.3 Outlier Analysis

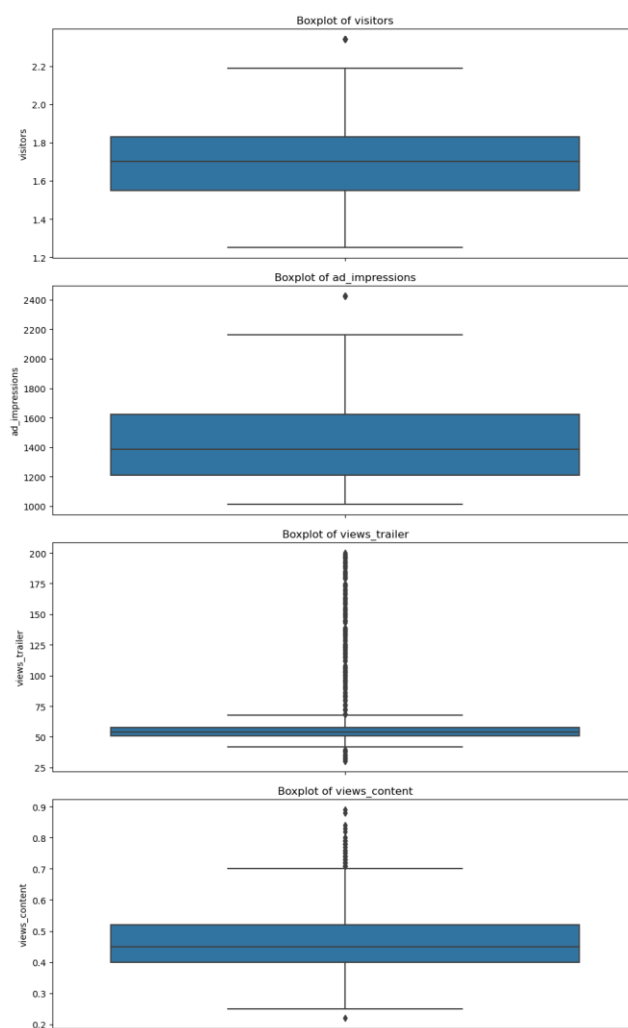


Figure 2: Boxplots of Numerical Variables — Outlier Screening

Variable	Potential Outliers (IQR)	Lower Bound	Upper Bound
visitors	20	1.13	2.25
ad_impressions	13	590.32	2243.68
views_trailer	189	40.74	67.97
views_content	47	0.22	0.70

Outlier Treatment Decision

The IQR method flags potential outliers in all four numerical variables, with `views_trailer` the most affected at 18.9% of observations. Inspection of the extreme values shows two distinct patterns: `visitors` and `ad_impressions` both have their top-ten values tied at exactly 2.34 and 2,424.20 respectively, indicating the source data is already capped at a ceiling rather than containing runaway errors. `views_trailer` and `views_content` decline gradually from their maxima, consistent with a genuine long right tail of high-performing content rather than data errors.

No observations are removed or capped. Removing them would discard exactly the high-viewership releases the business most wants to understand. Whether these observations exert undue influence on the fitted model is checked formally using Cook's distance after the model is built (Section 5.6).

2.4 Distribution of Genres (Question 2)

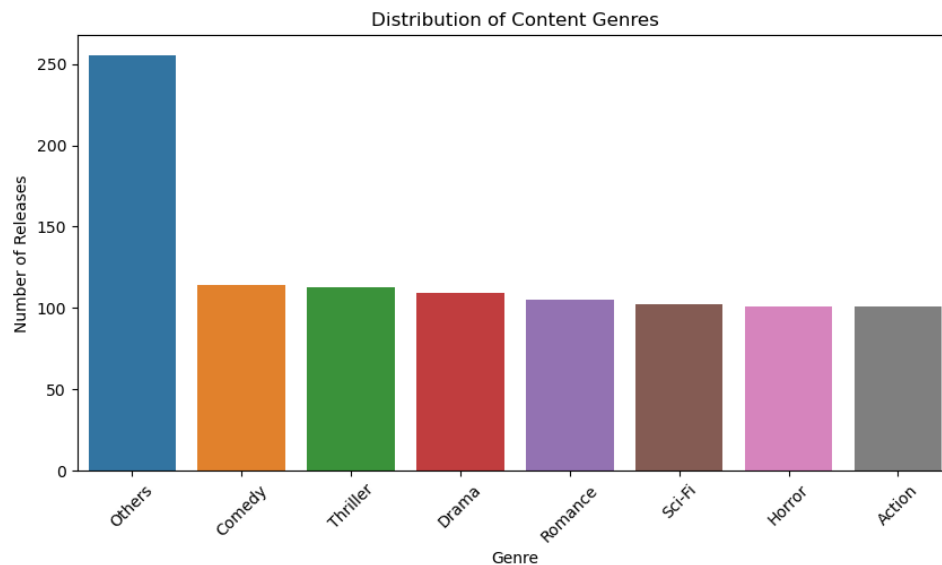


Figure 3: Distribution of Content Genres

Genre	Count
Others	255
Comedy	114
Thriller	113
Drama	109
Romance	105
Sci-Fi	102
Horror	101
Action	101

Table 2: Genre Distribution Counts

Answer — What does the distribution of genres look like?

The Others category is by far the most common genre, accounting for 255 of 1,000 releases (25.5%). The remaining seven named genres are fairly evenly represented, ranging from 101 releases (Action, Horror) to 114 (Comedy). No single named genre dominates the catalogue, and each has enough observations (100+) to support reliable comparison in the regression model.

2.5 Univariate Analysis of Remaining Categorical Variables

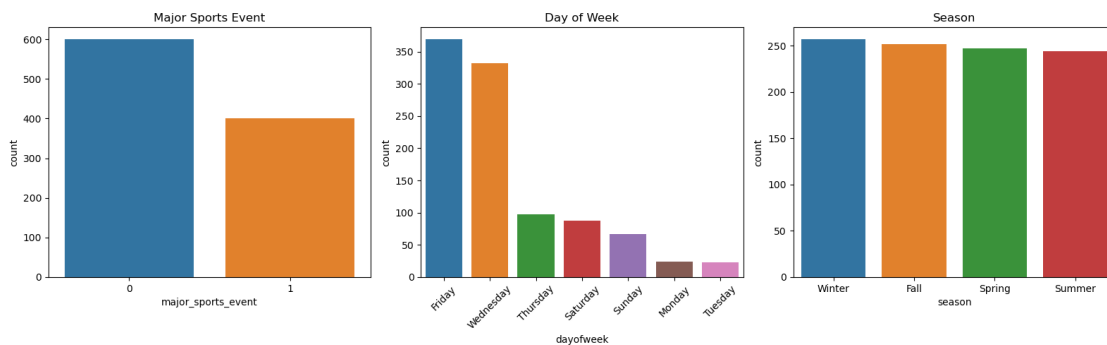


Figure 4: Distribution of Major Sports Event, Day of Week and Season

major_sports_event splits 60/40 (no event / event). Release-day frequency is highly imbalanced — Friday (369) and Wednesday (332) together account for 70% of all releases, while Monday (24) and Tuesday (23) are rarely used. Season is close to evenly split (244–257 per season). The heavy concentration on Friday and Wednesday, and the very small Monday/Tuesday samples, are important context for interpreting the day-of-week regression coefficients later.

2.6 Bivariate Analysis — Numerical Predictors vs Target



Figure 5: Numerical Predictors vs First-Day Content Views

Predictor	Correlation with views_content
views_trailer	0.754
visitors	0.259
ad_impressions	0.050

Table 3: Correlation of Numerical Predictors with First-Day Views

Answer — What is the correlation between trailer views and content views?

The Pearson correlation between views_trailer and views_content is 0.754, indicating a strong positive linear relationship — by far the strongest of any numerical predictor examined. Content with higher trailer viewership consistently achieves higher first-day views.

For comparison, visitors correlates at 0.259 (weak positive) and ad_impressions at 0.050 (effectively no linear relationship). Advertising volume, despite spanning more than a two-fold range across releases (1,011 to 2,424 million impressions), shows almost no association with first-day viewership on its own.

2.7 Bivariate Analysis — Day of Week (Question 3)

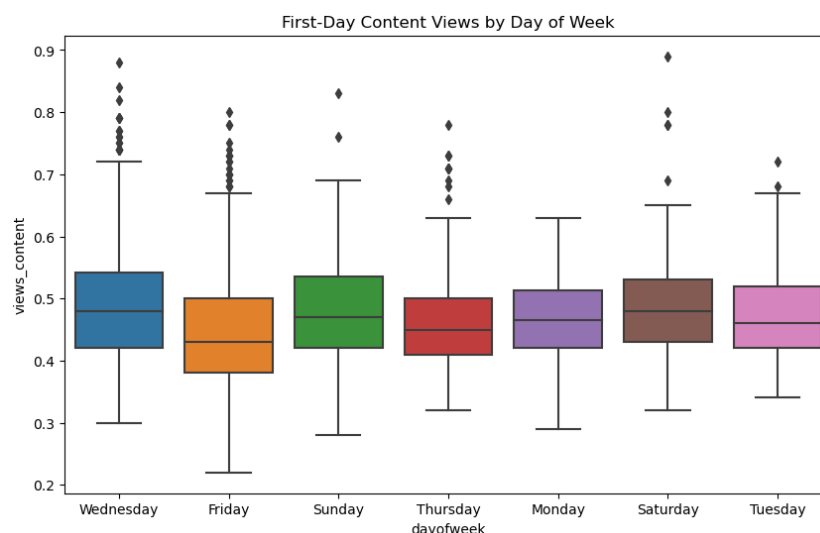


Figure 6: First-Day Content Views by Day of Week

Day	Count	Mean	Median	Std Dev
Saturday	88	0.498	0.480	0.101
Wednesday	332	0.495	0.480	0.108
Tuesday	23	0.488	0.460	0.110
Sunday	67	0.484	0.470	0.110
Thursday	97	0.471	0.450	0.101
Monday	24	0.468	0.465	0.083
Friday	369	0.447	0.430	0.102

Table 4: First-Day Views by Day of Week — Descriptive Statistics

Answer — How does viewership vary with the day of release?

Average first-day viewership varies meaningfully across release days. Saturday has the highest mean viewership (0.498 million), closely followed by Wednesday (0.495 million), while Friday has the lowest (0.447 million) — a gap of roughly 0.051 million views, or about 11% of Friday's own mean.

Friday is also the single most-used release day (369 of 1,000 releases), meaning the platform's heaviest release day is its weakest-performing one.

Monday and Tuesday show relatively high means but rest on very few observations (24 and 23) and should be interpreted cautiously. This relationship is tested formally, controlling for other factors, in the regression model.

2.8 Bivariate Analysis — Season (Question 4)

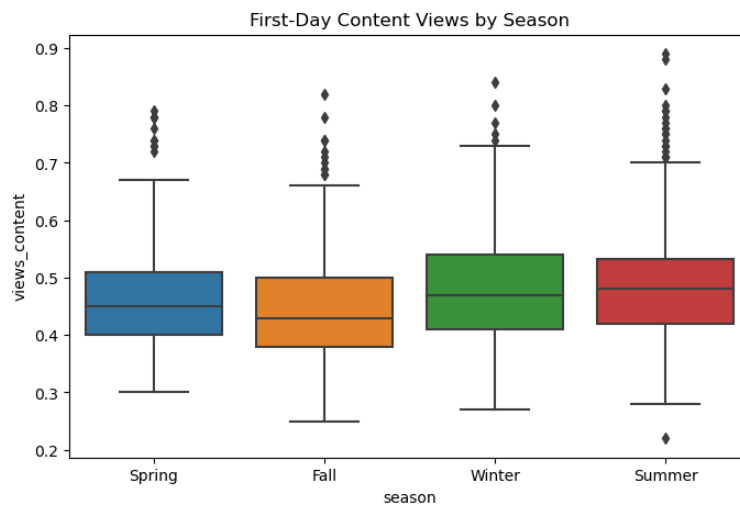


Figure 7: First-Day Content Views by Season

Season	Count	Mean	Median	Std Dev
Summer	244	0.497	0.48	0.112
Winter	257	0.485	0.47	0.106
Spring	247	0.467	0.45	0.098
Fall	252	0.445	0.43	0.102

Table 5: First-Day Views by Season — Descriptive Statistics

Answer — How does viewership vary with the season of release?

First-day viewership shows a consistent gradient across seasons: Summer is highest (0.497 million), followed by Winter (0.485 million), Spring (0.467 million), and Fall is lowest (0.445 million). Since the four seasons are almost evenly represented (244–257 releases each), this comparison is not distorted by sample-size imbalance, unlike the day-of-week comparison above. The gap between Summer and Fall (0.051 million, ~11%) is comparable in size to the Saturday-vs-Friday gap.

2.9 Bivariate Analysis — Major Sports Event

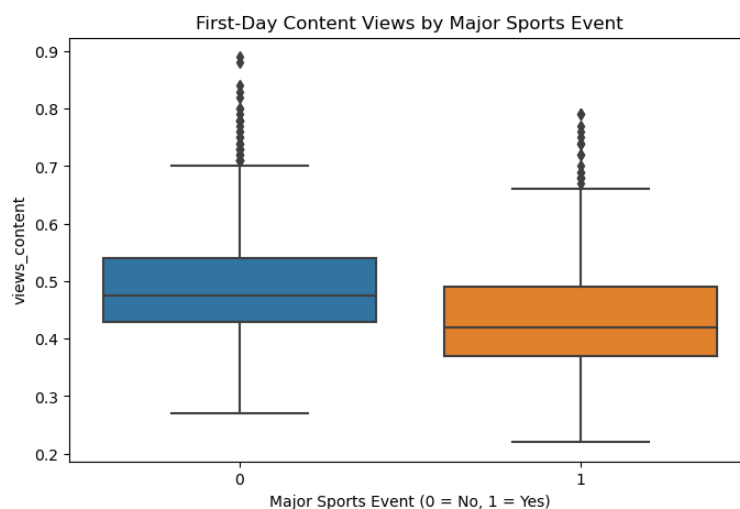


Figure 8: First-Day Content Views by Major Sports Event

Major Sports Event	Count	Mean	Median	Std Dev
No (0)	600	0.494	0.475	0.101
Yes (1)	400	0.443	0.420	0.105

Table 6: First-Day Views by Major Sports Event — Descriptive Statistics

Content released without a competing major sports event averages 0.494 million first-day views versus 0.443 million when a major event occurs — a gap of roughly 0.051 million (10.3%). This suggests sports events compete for viewer attention on release day.

2.10 Bivariate Analysis — Genre

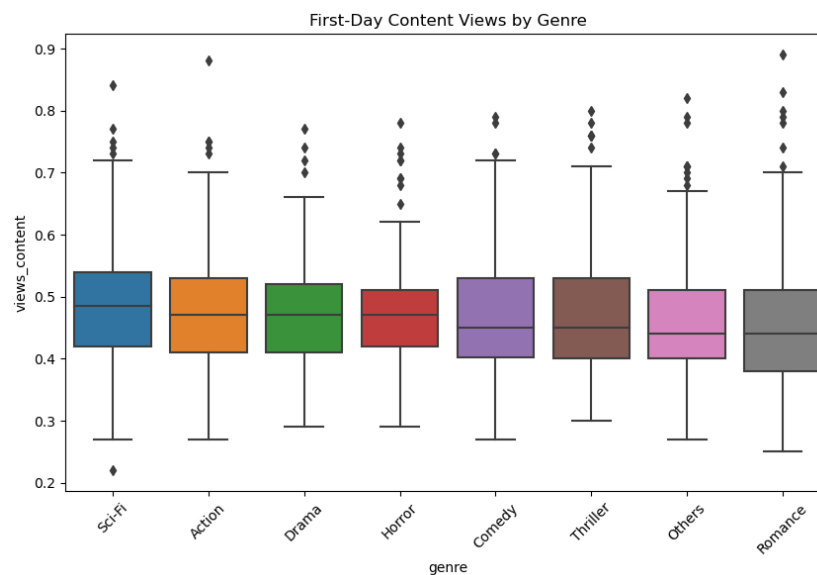


Figure 9: First-Day Content Views by Genre

Genre	Count	Mean	Median	Std Dev
Sci-Fi	102	0.493	0.485	0.113
Action	101	0.487	0.470	0.111
Horror	101	0.481	0.470	0.098
Thriller	113	0.477	0.450	0.114
Comedy	114	0.474	0.450	0.103
Drama	109	0.474	0.470	0.096
Romance	105	0.461	0.440	0.123
Others	255	0.460	0.440	0.097

Table 7: First-Day Views by Genre — Descriptive Statistics

Genre means span only 0.460 to 0.493 million views — a much narrower spread than day-of-week or season — with substantial overlap between distributions. Sci-Fi and Action are highest, Others and Romance lowest, but the

differences are small relative to the within-genre variation. Genre's independent contribution is tested formally in the regression model.

2.11 Multivariate Analysis

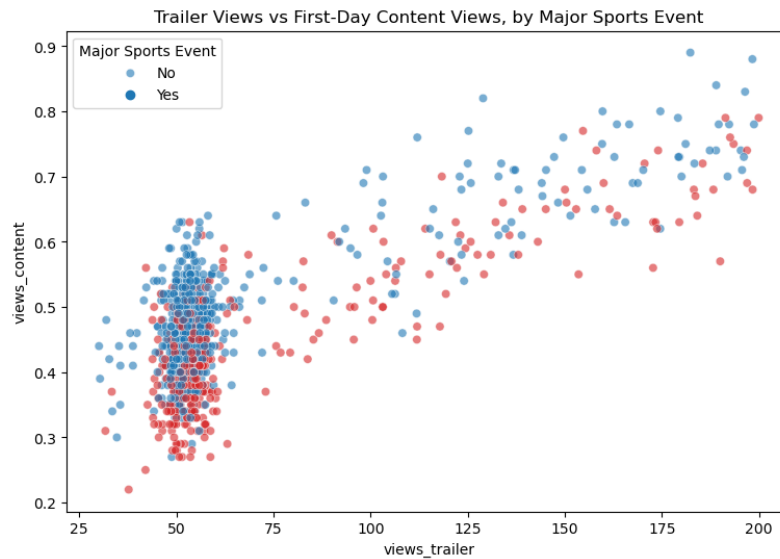


Figure 10: Trailer Views vs Content Views, by Major Sports Event

The positive trailer-views relationship holds within both groups, but releases coinciding with a sporting event sit consistently below those without one at comparable trailer-view levels. This indicates the sports-event penalty operates largely independently of trailer engagement, rather than being an artefact of weaker content happening to be scheduled on those days.

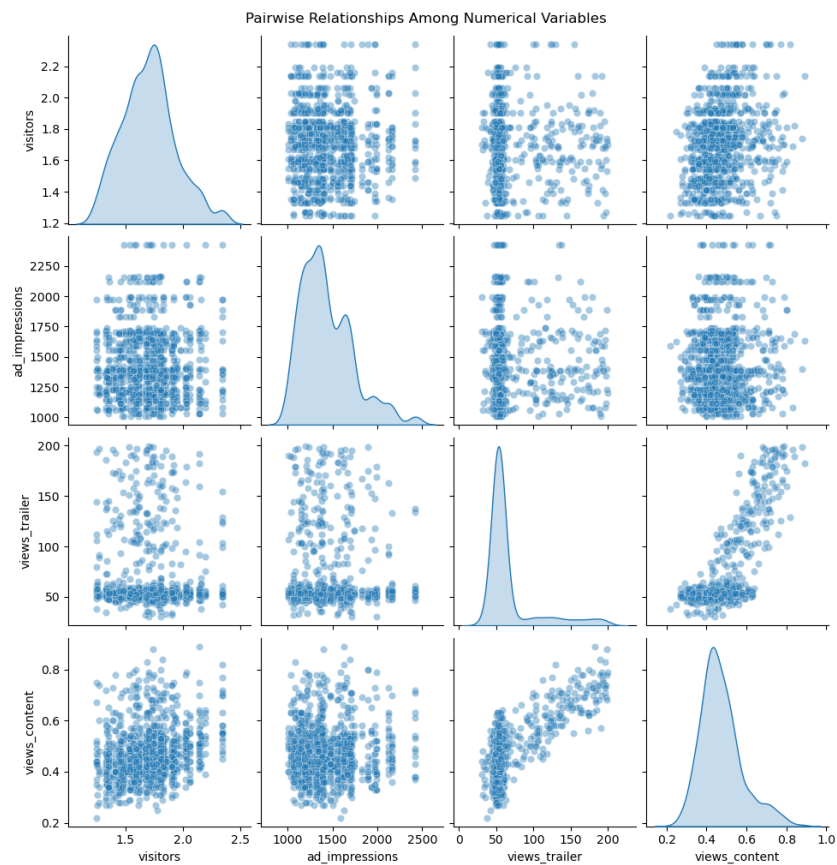


Figure 11: Pairwise Relationships Among Numerical Variables

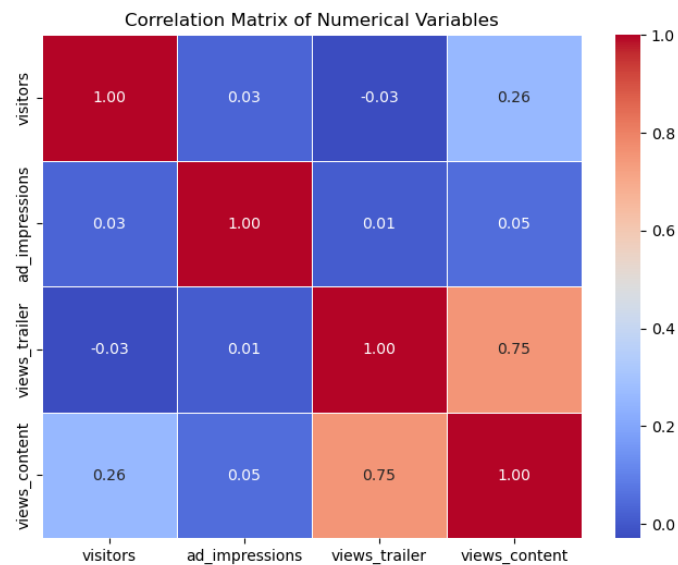


Figure 12: Correlation Matrix of Numerical Variables

The three numerical predictors are essentially uncorrelated with one another (all $|r| < 0.04$), which is favourable for regression — their effects on `views_content` can be estimated independently without multicollinearity concerns among them.

2.12 Summary of EDA Insights

- Target variable: first-day viewership averages 0.473 million views (median 0.45), positively skewed, ranging 0.22–0.89 million.
- Strongest driver: views_trailer ($r = 0.754$) — far stronger than visitors ($r = 0.259$) or ad_impressions ($r = 0.050$).
- Advertising volume shows almost no linear relationship with viewership despite a two-fold range across releases.
- Sporting events depress viewership by roughly 10% on average.
- Release day is the clearest operational finding: Friday is used most (37% of releases) but performs worst; Saturday and Wednesday perform best.
- Season shows a consistent gradient: Summer > Winter > Spring > Fall.
- Genre shows limited separation between categories relative to day/season effects.
- Numerical predictors are nearly uncorrelated with each other, which is favourable for the regression model that follows.

Section 3 — Data Preprocessing

3.1 Feature Engineering

Three candidate features were constructed and tested against their raw counterparts. A feature is retained only if there is evidence it improves the model: a log transform of trailer views (to address its high skewness), an advertising-intensity ratio (ad impressions per visitor), and a weekend flag (collapsing the seven release days into weekday/weekend).

Feature	Correlation with views_content	Skewness
views_trailer (raw)	0.7540	2.3729
log_views_trailer	0.7512	1.8655
ad_impressions (raw)	0.0500	1.0336
ad_per_visitor	-0.0986	0.8240

Table 8: Feature Engineering — Candidate Feature Comparison

Feature Retention Decision

- **log_views_trailer — not retained.** The log transform reduces skewness from 2.37 to 1.87 but slightly weakens the correlation with the target (0.751 vs 0.754 raw). Since the untransformed relationship is already strongly linear and later diagnostics show no evidence of non-linearity, the raw variable is kept for interpretability — its coefficient reads directly as “additional first-day views per million trailer views”.
- **ad_per_visitor — not retained.** Advertising intensity correlates at -0.099, which is larger in magnitude than raw impressions (0.050) but now negative — more ad exposure relative to platform size is weakly associated with lower viewership. This is more likely a proxy for smaller-audience releases receiving heavier relative promotion than a genuine effect, and does not offer a cleaner story than the raw variable.
- **is_weekend — not retained.** Weekend releases do average higher viewership (0.492 vs 0.470), but the flag would misclassify the most important pattern in the data: Wednesday (0.4946) performs almost identically to the weekend average, while Friday (0.4467) — a weekday — is the worst performer of all. Collapsing seven days into two would erase this distinction. The full day-of-week dummy encoding is retained; the thin Monday/Tuesday samples are flagged as a limitation.

Conclusion: no transformation is applied beyond categorical dummy encoding. The engineered features were constructed and tested; the raw specification is retained because the evidence does not support replacing it.

3.2 Encoding Categorical Variables

The three categorical predictors — genre, dayofweek, and season — were one-hot encoded using `drop_first=True` to avoid the dummy-variable trap. This expanded the predictor set from 7 raw columns to 20 encoded columns (1,000 rows × 20 columns after encoding, excluding the target).

Categorical Variable	Reference (Omitted) Category
genre	Action
dayofweek	Friday
season	Fall

Table 9: Reference (Omitted) Categories After Dummy Encoding

Every dummy coefficient in the model is interpreted relative to these reference categories — for example, the `dayofweek_Saturday` coefficient is read as the difference in first-day views between a Saturday release and an otherwise-identical Friday release.

3.3 Train-Test Split

The dataset was split into training and testing sets using an 80/20 split (`random_state = 42`), so the model's ability to generalise to unseen releases can be evaluated honestly rather than only on the data it was fitted to.

Training set: 800 releases × 20 predictors

Testing set: 200 releases × 20 predictors

No missing values or non-numeric columns present in either split — data is model-ready.

Section 4 — Linear Regression Model

4.1 Model Building

A multiple linear regression model was fitted on the training data (800 releases) using platform traffic, advertising, trailer engagement, release timing, season, genre, and the sports-event indicator as predictors, using Ordinary Least Squares (OLS) via statsmodels.

4.2 Overall Model Statistics

Statistic	Value
R-squared	0.7868
Adjusted R-squared	0.7814
F-statistic	143.77
F-statistic p-value	2.53×10^{-245}
AIC	-2516.03
BIC	-2417.66
No. of observations (training)	800
Df. Model	20

4.3 Coefficient Summary

Predictor	Coefficient	Std Error	t-stat	p-value
const	0.0548	0.0176	3.11	0.002
visitors	0.1289	0.0076	16.87	< 0.001
ad_impressions	0.0000081	0.0000061	1.34	0.180
major_sports_event	-0.0596	0.0037	-16.28	< 0.001
views_trailer	0.0023	0.00005	45.85	< 0.001
genre_Comedy	0.0044	0.0078	0.56	0.573
genre_Drama	0.0106	0.0077	1.37	0.170
genre_Horror	0.0094	0.0078	1.21	0.225
genre_Others	0.0050	0.0067	0.74	0.459
genre_Romance	-0.0014	0.0077	-0.18	0.858
genre_Sci-Fi	0.0100	0.0081	1.24	0.215
genre_Thriller	0.0115	0.0078	1.48	0.140
dayofweek_Monday	0.0451	0.0121	3.71	< 0.001
dayofweek_Saturday	0.0526	0.0068	7.69	< 0.001
dayofweek_Sunday	0.0388	0.0073	5.29	< 0.001
dayofweek_Thursday	0.0196	0.0062	3.18	0.002
dayofweek_Tuesday	0.0324	0.0125	2.59	0.010

Predictor	Coefficient	Std Error	t-stat	p-value
dayofweek_Wednesday	0.0495	0.0042	11.71	< 0.001
season_Spring	0.0232	0.0050	4.63	< 0.001
season_Summer	0.0446	0.0051	8.71	< 0.001
season_Winter	0.0265	0.0050	5.30	< 0.001

Table 10: Full Model — Coefficient Summary

4.4 Statistically Significant Predictors

Predictor	Coefficient	p-value	Interpretation
visitors	0.1289	< 0.001	Positive — more platform traffic drives more views
major_sports_event	−0.0596	< 0.001	Negative — a sporting-event clash reduces views
views_trailer	0.0023	< 0.001	Positive — strongest predictor in the model
All 6 dayofweek dummies	0.020 to 0.053	< 0.05 (all)	Every day outperforms Friday (reference)
All 3 season dummies	0.023 to 0.045	< 0.001 (all)	Every season outperforms Fall (reference)

Table 11: Statistically Significant Predictors ($p < 0.05$)

The model explains 78.68% of the variation in first-day viewership on training data (adjusted $R^2 = 78.14\%$). The overall F-test is significant ($p < 0.001$). visitors, major_sports_event, and views_trailer are significant numerical/binary predictors; visitors and trailer views are positively associated with viewership, while sports events are negatively associated. All day-of-week and season dummies are significant relative to their Friday and Fall reference categories, all with positive coefficients. ad_impressions and all genre dummies are not statistically significant at the 5% level.

4.5 Reduced Model — Robustness Check

To confirm that ad_impressions and the seven genre dummies can be safely reported as non-significant rather than merely individually insignificant, a reduced model dropping all eight of these predictors was fitted and compared to the full model using a partial F-test.

Model	R-squared	Adj. R-squared	AIC	BIC
Full model (20 predictors)	0.7868	0.7814	−2516.03	−2417.66
Reduced model (12 predictors)	0.7846	0.7813	−2523.69	−2462.79

Table 12: Full vs Reduced Model Comparison

Partial F-test: $F = 1.0210$, $p = 0.4183$, $df = 8$

Since $p > 0.05$, the eight dropped coefficients (ad_impressions and the seven genre dummies) are not jointly significant — dropping them costs no explanatory power. The reduced model's adjusted R^2 (0.7813) is essentially identical to the full model's (0.7814), while both its AIC and BIC are lower, favouring the simpler model on parsimony grounds.

Decision: the full model is retained as the primary specification, since it lets the non-significance of advertising and genre be reported explicitly as a business finding. The reduced model confirms this choice costs nothing in fit.

Section 5 — Testing Assumptions of Linear Regression

The validity of a linear regression model's coefficients and inference depends on five classical assumptions: linearity, independence of errors, normality of residuals, homoscedasticity, and absence of problematic multicollinearity. Each is tested formally below, along with an influence diagnostic to check that no single observation drives the fitted coefficients.

5.1 Linearity

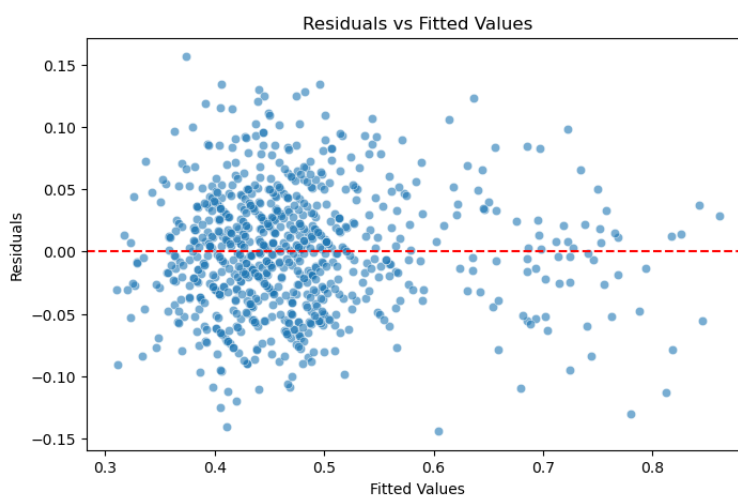


Figure 13: Residuals vs Fitted Values



Figure 14: Residuals vs Each Continuous Predictor

Mean of residuals: 7.80×10^{-14} (effectively zero, as expected for OLS with an intercept)

Finding

The residuals are scattered around zero with no systematic curved pattern, either against fitted values or against each continuous predictor individually. The linearity assumption is reasonably satisfied.

5.2 Independence of Errors

Durbin-Watson statistic: 2.038

Finding

The Durbin-Watson statistic (2.038) is close to 2, indicating no evidence of first-order autocorrelation in the residuals. The independence assumption is satisfied.

5.3 Normality of Residuals

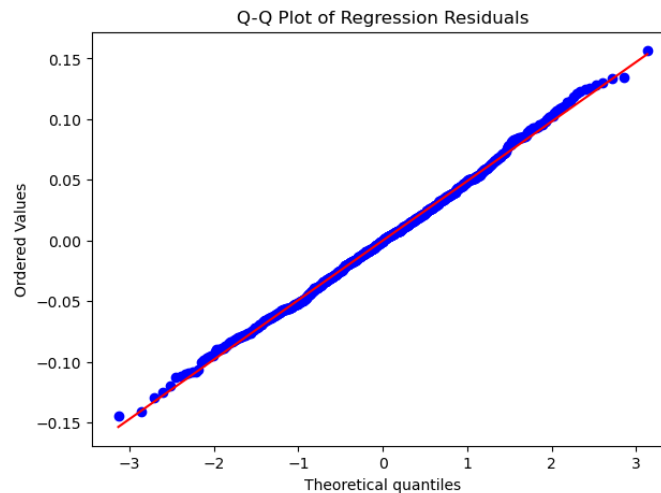


Figure 15: Q-Q Plot of Regression Residuals

Shapiro-Wilk statistic: 0.9979

Shapiro-Wilk p-value: 0.4376

Finding

The Q-Q plot shows the residuals closely follow the theoretical normal line, with only minor tail deviations. The Shapiro-Wilk p-value (0.4376) exceeds 0.05, so we fail to reject the null hypothesis of normality. The normality assumption is reasonably satisfied.

5.4 Homoscedasticity

Breusch-Pagan statistic: 20.542

Breusch-Pagan p-value: 0.4245

Finding

The Breusch-Pagan p-value (0.4245) exceeds 0.05, so we fail to reject the null of constant residual variance. There is no significant evidence of heteroscedasticity; the homoscedasticity assumption is reasonably satisfied.

5.5 Multicollinearity

Predictor	VIF
genre_Others	2.75

Predictor	VIF
genre_Horror	1.96
genre_Thriller	1.95
genre_Romance	1.95
genre_Drama	1.95
genre_Comedy	1.90
genre_Sci-Fi	1.84
season_Summer	1.57
season_Winter	1.56
season_Spring	1.54
dayofweek_Wednesday	1.29
dayofweek_Thursday	1.17
dayofweek_Saturday	1.13
dayofweek_Sunday	1.12
dayofweek_Tuesday	1.06
dayofweek_Monday	1.06
major_sports_event	1.05
visitors	1.02
views_trailer	1.02
ad_impressions	1.02

Table 13: Variance Inflation Factors (VIF)

Finding

Excluding the intercept (whose high VIF of 101.1 is expected and not meaningful — it reflects that predictors are not centred on zero, not multicollinearity), all predictors have VIF well below 5, with the highest being 2.75 (genre_Others). There is no evidence of problematic multicollinearity.

5.6 Influence Analysis

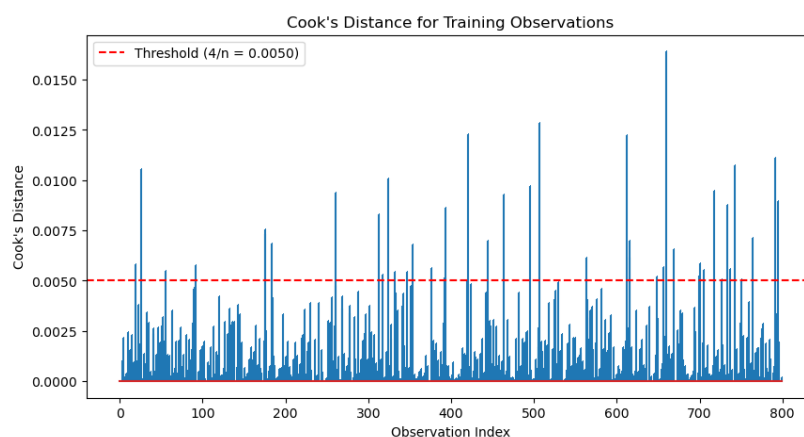


Figure 16: Cook's Distance for Training Observations

Observations above 4/n threshold: 43

Observations with Cook's D > 0.5: 0

Maximum Cook's Distance: 0.0164

Finding

The maximum Cook's distance is 0.0164, far below the conventional highly-influential threshold of 0.5. This confirms the decision made during preprocessing to retain all observations without capping or removal: the high-viewership outliers identified during EDA are genuine business observations and do not exert undue leverage on the fitted coefficients.

5.7 Summary of Assumption Checks

Assumption	Test Used	Result	Satisfied?
Linearity	Residuals vs Fitted / Predictors	No systematic pattern	Yes
Independence of errors	Durbin-Watson	DW = 2.038	Yes
Normality of residuals	Shapiro-Wilk, Q-Q Plot	p = 0.4376	Yes
Homoscedasticity	Breusch-Pagan	p = 0.4245	Yes
No problematic multicollinearity	VIF	Max VIF = 2.75	Yes
No high-influence observations	Cook's Distance	Max D = 0.0164	Yes

Table 14: Summary of Regression Assumption Checks

All five classical regression assumptions — plus the influence diagnostic — are satisfied. The model's coefficients and p-values can be interpreted with confidence.

Section 6 — Model Performance Evaluation

6.1 Performance Metrics

Metric	Training	Testing
MAE (million views)	0.0389	0.0399
RMSE (million views)	0.0489	0.0500
MAPE (%)	8.61	9.08
R-squared	0.7868	0.7743

Table 15: Model Performance Metrics — Training vs Testing

6.2 Generalisation Check

Training R-squared: 0.7868

Testing R-squared: 0.7743

Difference: 0.0125

RMSE as % of mean viewership: 10.71%

The model explains 78.68% of variation on training data and 77.43% on unseen test data — a gap of only 0.0125, indicating the model generalises well rather than overfitting. The test-set mean absolute error is 0.0399 million views and RMSE is 0.0500 million, which is 10.71% of the average viewership (0.473 million). The mean absolute percentage error is 9.08%.

RMSE exceeds MAE because it penalises large errors more heavily, but the gap between them is modest — the model has no small set of severely mispredicted releases. In practical terms, a typical release's first-day forecast lands within roughly 40,000 views of the actual figure — precise enough to support scheduling and capacity-planning decisions, though not to forecast an individual release exactly.

6.3 Actual vs Predicted Viewership

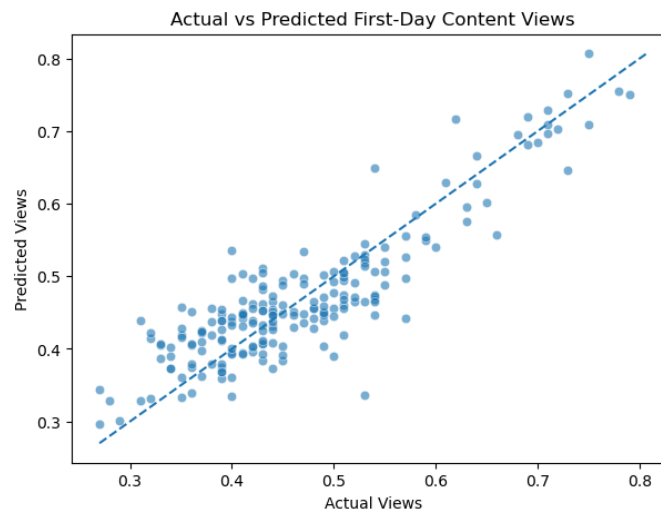


Figure 17: Actual vs Predicted First-Day Content Views

Predicted values track actual values closely along the diagonal reference line across the full range of viewership, with no evidence of systematic under- or over-prediction at either the low or high end. This visual pattern corroborates the test-set R^2 of 0.774 and confirms the model performs consistently across the range of release outcomes, not just on average.

Section 7 — Actionable Business Insights and Recommendations

7.1 Key Business Insights

1. Trailer engagement is the single strongest lever on first-day viewership

views_trailer has coefficient 0.002311 ($p < 0.001$, $t = 45.8$) — the most significant predictor in the model. Every additional 1 million trailer views is associated with roughly 2,311 additional first-day content views. Across the observed trailer-performance range (30M to 200M views), this accounts for a swing of approximately 0.39 million first-day views — larger than the entire average viewership of 0.473 million.

2. Releasing during a major sporting event costs roughly one-eighth of the audience

major_sports_event = -0.059559 ($p < 0.001$). A release coinciding with a major sporting event loses approximately 59,600 first-day views — 12.6% of average viewership — after controlling for trailer engagement, traffic, genre, and season.

3. ShowTime releases most of its content on its worst-performing day

Friday carries 369 of 1,000 releases (37%) and has the lowest mean first-day viewership of any day (0.447 million). Every other day carries a positive, significant coefficient against Friday:

Release Day	Coefficient vs Friday	Additional First-Day Views	Releases in Data
Saturday	+0.0526	~52,600	88
Wednesday	+0.0495	~49,500	332
Monday	+0.0451	~45,100	24
Sunday	+0.0388	~38,800	67
Tuesday	+0.0324	~32,400	23
Thursday	+0.0196	~19,600	97
Friday	reference	—	369

Table 16: Release-Day Effect on First-Day Viewership (vs Friday)

Saturday outperforms Friday by ~52,600 views (11.1% of average), yet is used for only 88 releases. Wednesday performs almost as well and is already used heavily (332 releases), confirming the platform can execute mid-week releases successfully.

Caution: Monday and Tuesday rest on only 24 and 23 observations and carry wide confidence intervals — Saturday and Wednesday are the well-supported estimates to act on.

4. Platform traffic contributes a meaningful but smaller effect

visitors = 0.128909 ($p < 0.001$): each additional million weekly platform visitors is associated with ~12,900 additional first-day views. Moving from the 25th to 75th percentile of traffic (1.55M → 1.83M) is worth ~36,100 views, or 7.6% of average viewership.

5. Season produces a consistent gradient, Summer strongest

Season	Coefficient vs Fall	Additional First-Day Views	% of Mean
Summer	+0.0446	~44,600	9.4%
Winter	+0.0265	~26,500	5.6%
Spring	+0.0232	~23,200	4.9%
Fall	reference	—	—

Table 17: Season Effect on First-Day Viewership (vs Fall)

6. Advertising volume shows no independent effect

ad_impressions is not significant ($p = 0.180$); raw correlation with viewership is 0.050. Across the observed range (1,011–2,424 million impressions), the model attributes a difference of only ~11,000 views — not statistically distinguishable from zero.

7. Genre does not independently predict first-day viewership

None of the seven genre dummies is significant ($p \geq 0.14$). Once trailer engagement, timing and traffic are controlled for, genre adds nothing measurable.

Combined Effect

A release moved from Friday to Saturday (+52,600), scheduled in Summer instead of Fall (+44,600), and kept clear of a major sporting event (+59,600) gains approximately **156,800 first-day views — a 33% improvement over average viewership** — with no change to the content itself or the marketing budget.

7.2 Business Recommendations

- **1. Rebalance the release calendar away from Friday — highest priority.** Friday absorbs 37% of releases while performing worst. Shift high-value titles toward Saturday and Wednesday; even reallocating 50 releases a year could add ~2.5 million first-day views annually. Validate with a controlled test before restructuring the full calendar, since the Friday concentration may reflect contractual or industry norms not captured here.
- **2. Screen release dates against the major sporting calendar.** A clash costs ~12.6% of first-day viewership and is largely avoidable, since sporting calendars are published well in advance.
- **3. Use trailer performance as a pre-release forecasting instrument.** Trailer views are observable before release and are the strongest predictor available — forecast expected first-day demand at ~2,311 views per million trailer views to support capacity and promotional decisions.
- **4. Audit advertising effectiveness rather than advertising volume.** Impressions show no measurable link to first-day viewership across a two-fold spending range. Investigate targeting, creative, and channel mix; consider redirecting some impression spend toward trailer promotion.
- **5. Concentrate flagship releases in Summer; support Fall releases more heavily.** Summer outperforms Fall by ~9.4% of average viewership.

- **6. Stop using genre as a demand-planning input.** Genre carries no independent predictive power once other factors are controlled for.

Priority	Action	Est. Effect	Cost to Implement
1	Avoid sporting-event clashes	~59,600 views (12.6%)	Minimal
2	Shift Friday releases to Sat/Wed	~50,000 views (10.6%)	Low
3	Favour Summer over Fall windows	~44,600 views (9.4%)	Low
4	Trailer-based forecasting	Operational	Medium
5	Advertising effectiveness audit	TBD	Medium

Table 18: Prioritised Business Recommendations

Section 8 — Final Conclusion

8.1 Conclusion

The objective of this analysis was to identify the factors associated with first-day content viewership on the ShowTime platform and to develop a model to predict first-day views. A multiple linear regression model using platform visitors, advertising impressions, major sports event status, trailer views, genre, day of week, and season achieved an R^2 of 0.7868 (adjusted 0.7814) on training data and 0.7743 on unseen test data, with an F-test significant at $p < 0.001$.

visitors, views_trailer, major_sports_event, release day, and season were statistically significant predictors; advertising impressions and genre were not. All five regression assumptions were tested with no material violations: linearity held, Durbin-Watson = 2.038, Shapiro-Wilk $p = 0.4376$, Breusch-Pagan $p = 0.4245$, and all VIFs were below 5. Influence diagnostics confirmed no observation exerts outsized leverage on the fit.

8.2 Limitations

- **Association, not causation.** Releases were not randomly assigned to days, seasons, or sporting-event windows; scheduling may correlate with unobserved factors. Recommendations should be validated with controlled tests before being adopted at scale.
- **Thin support for two release days.** Monday ($n = 24$) and Tuesday ($n = 23$) carry wide confidence intervals and should not drive scheduling decisions on their own.
- **Unbalanced release-day distribution.** Friday and Wednesday account for ~70% of releases; the Friday finding is well-supported, sparser-day comparisons less so.
- **Omitted variables.** Production budget, cast, critical reception, prior-season performance, competing platforms, and audience demographics are not in the dataset and likely explain much of the remaining ~22% of unexplained variation.
- **Single-day outcome.** The target measures only first-day viewership; content that builds an audience over subsequent weeks is not captured.
- **Reference-category dependence.** All categorical coefficients are relative to Action (genre), Friday (day), and Fall (season).
- **Sporting-event variable is binary.** It does not distinguish a major final from a routine fixture.
- **Advertising measured as volume only.** The null result should be read as “impression volume does not predict first-day viewership,” not “advertising does not work.”

8.3 Executive Summary Recap

Area	Result
Business Objective	Identify factors driving first-day viewership and build a predictive model
Strongest Predictor	Trailer views ($r = 0.754$), positively associated with viewership
Traffic Effect	Platform visitors have a significant positive association

Area	Result
Sports Event Effect	Reduces estimated first-day viewership by ~12.6%
Day / Season Effect	Significant relative to reference categories; Friday (most-used day) performs worst
Non-significant Factors	Advertising impressions and genre
Training R^2 / Adj. R^2	0.7868 / 0.7814
Testing R^2 / MAE / RMSE / MAPE	0.7743 / 0.0399 / 0.0500 / 9.08%
Assumptions	All satisfied — linearity, independence (DW 2.038), normality (Shapiro p 0.4376), homoscedasticity (BP p 0.4245), no multicollinearity (max VIF 2.75), no high-influence points (max Cook's D 0.016)

Business Implication: ShowTime can use trailer engagement, platform traffic, release-day timing, seasonal patterns, and the sporting-events calendar as concrete, actionable inputs for release planning and first-day demand forecasting — with rebalancing away from Friday releases as the single highest-leverage, lowest-cost change available.